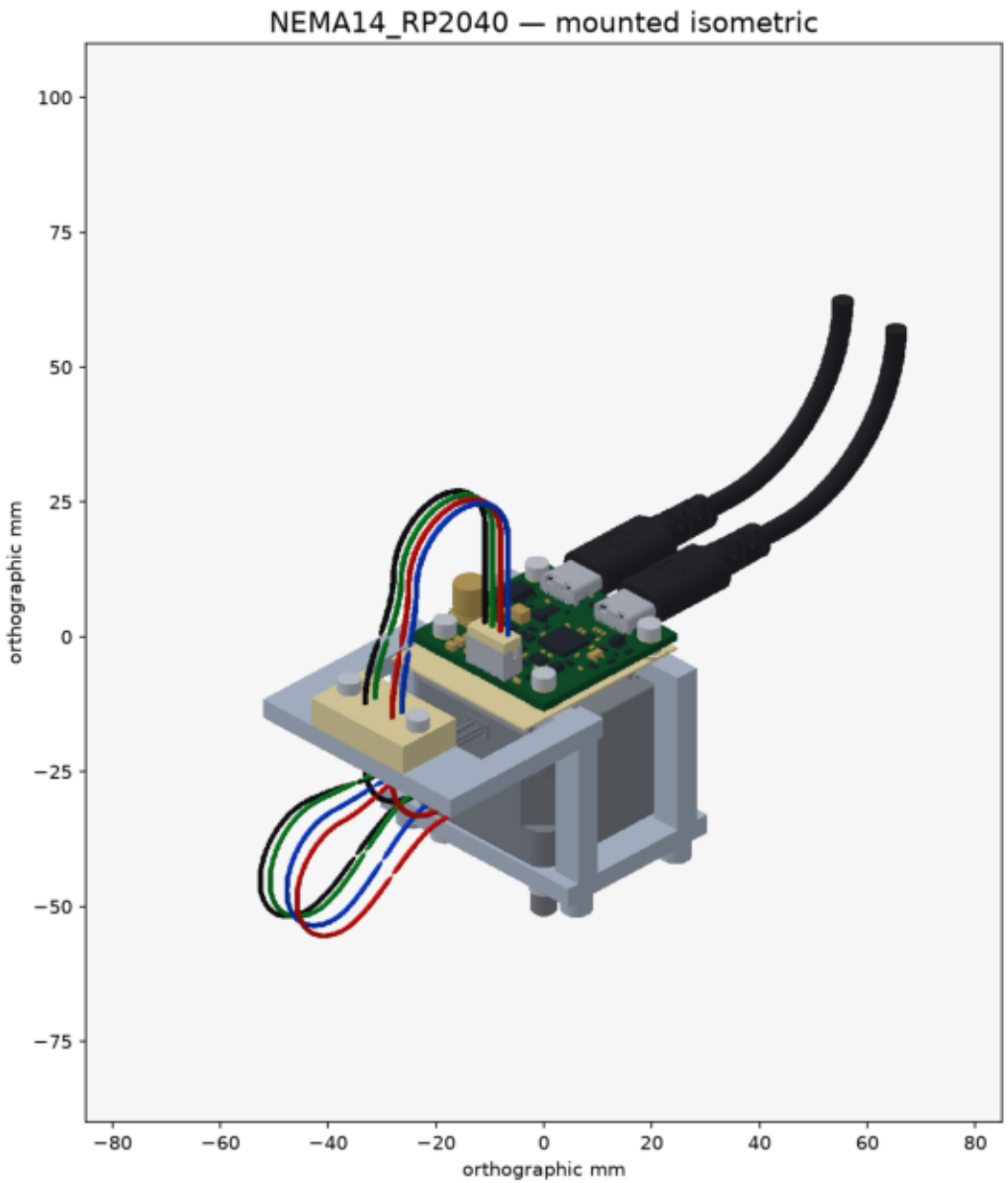
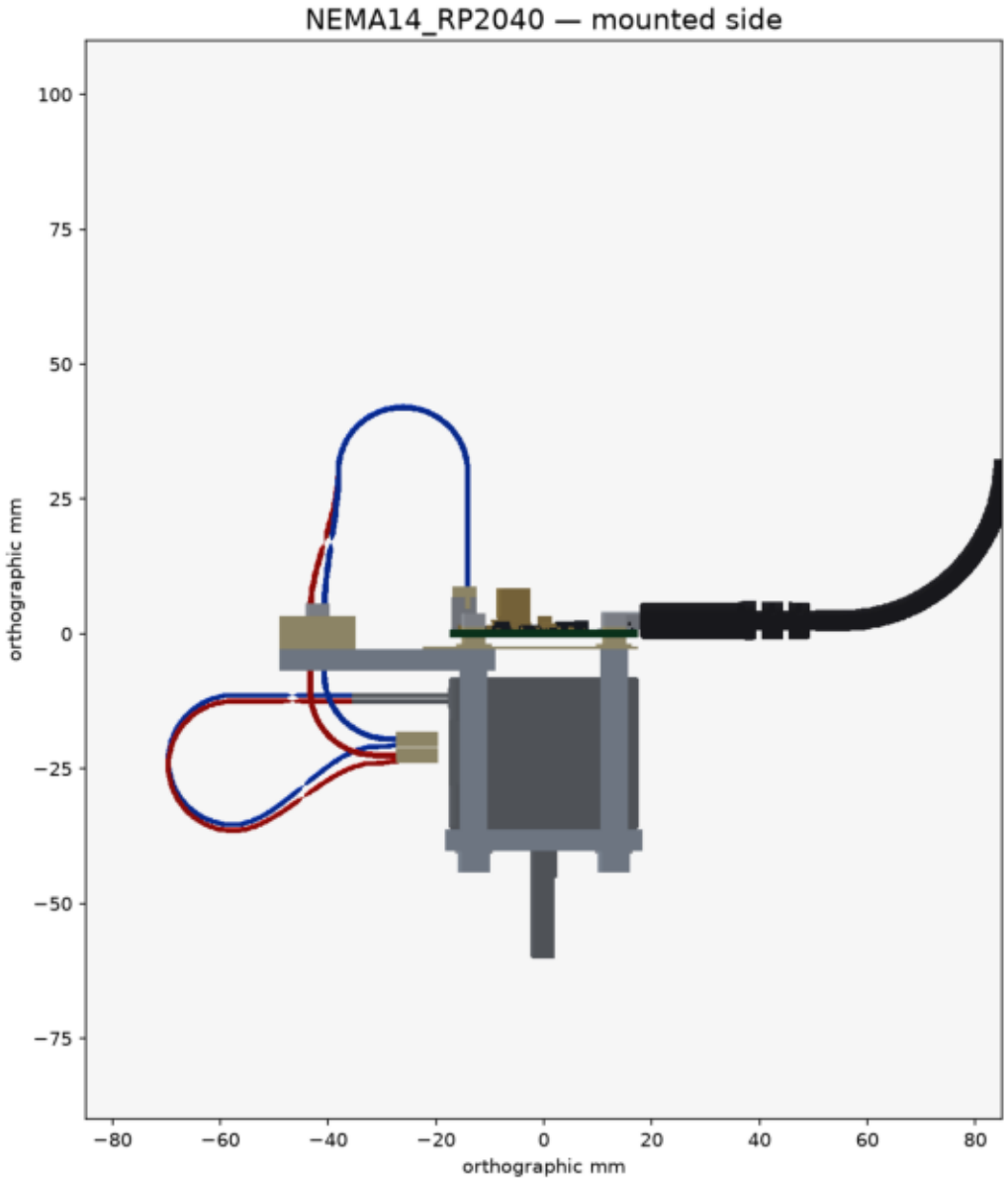
NEMA14_RP2040 — HAR-001 / HAR-002 installation



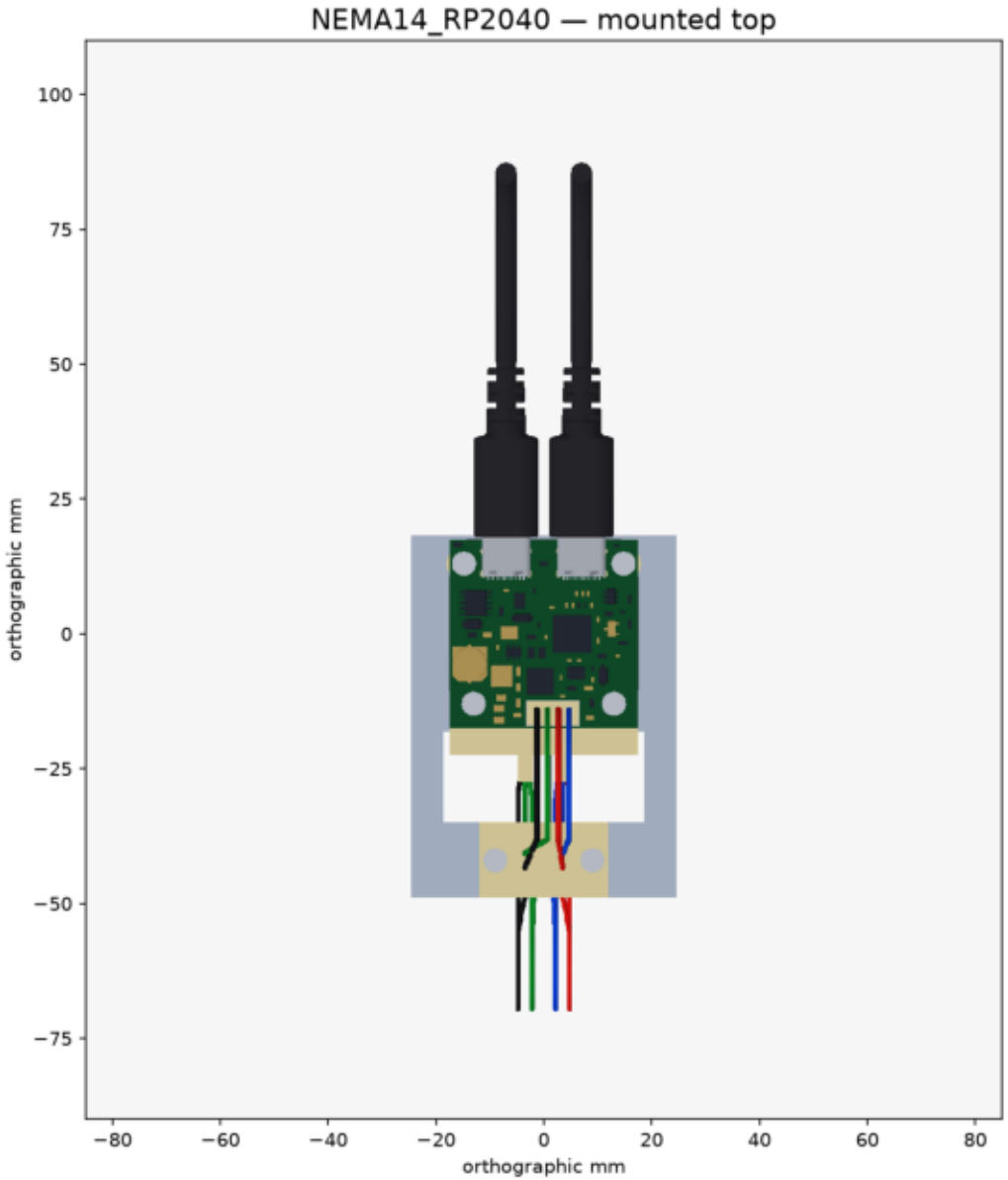
Nominal installation envelopes; factory key/contact and lead stock qualification pending. No measured fit.

NEMA14_RP2040 — HAR-001 / HAR-002 installation



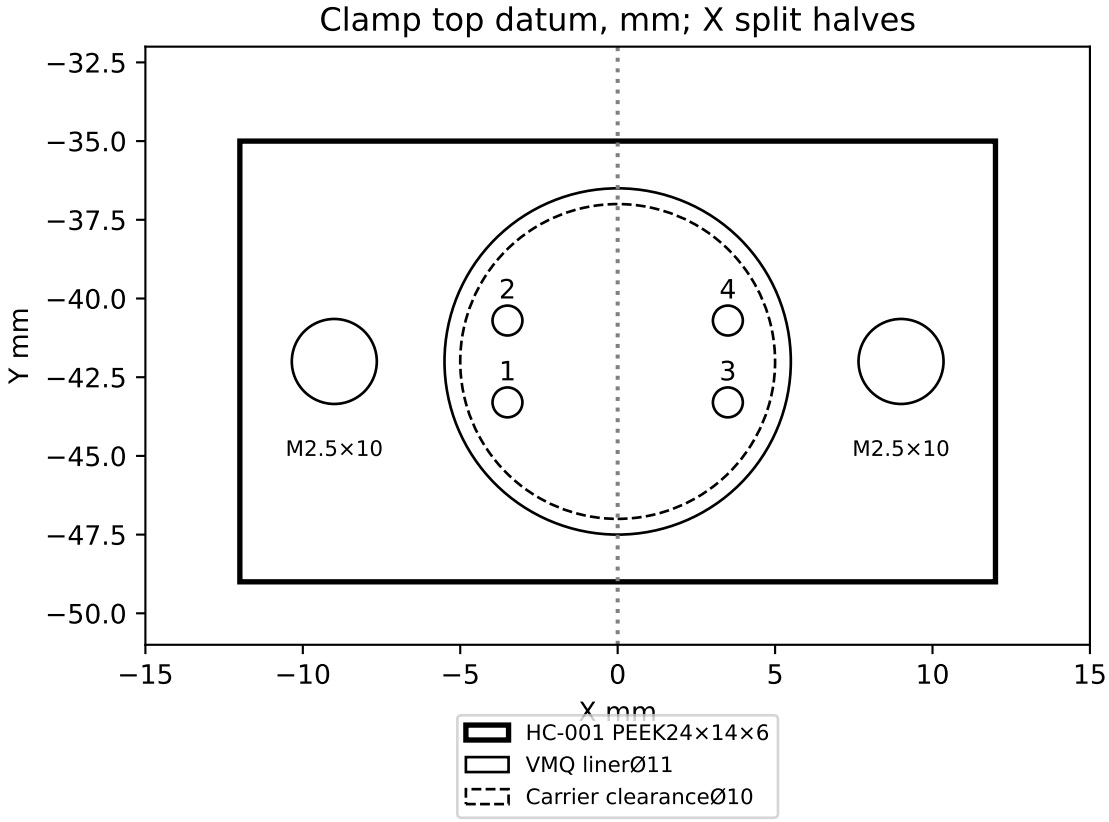
Nominal installation envelopes; factory key/contact and lead stock qualification pending. No measured fit.

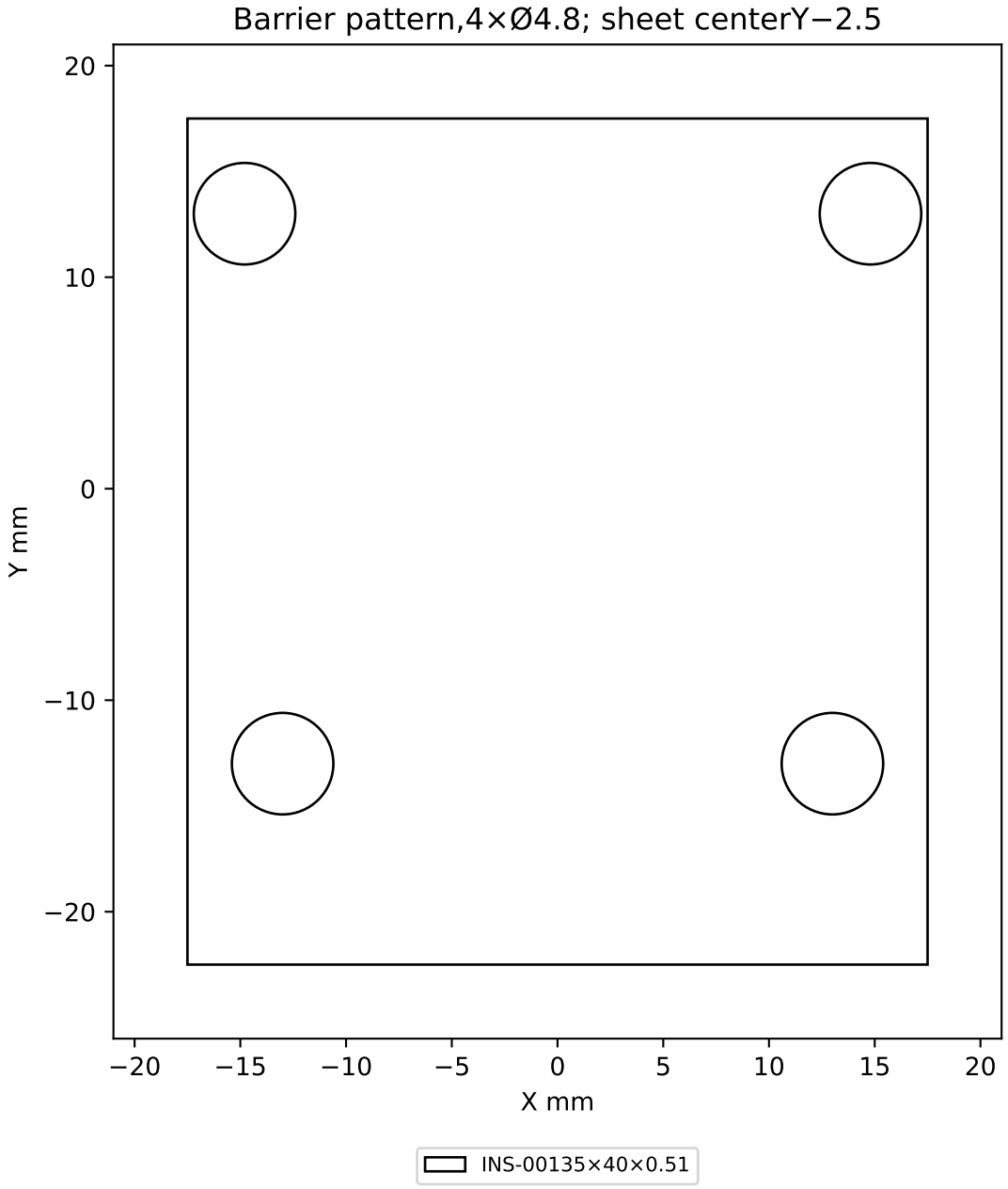
NEMA14_RP2040 — HAR-001 / HAR-002 installation



Nominal installation envelopes; factory key/contact and lead stock qualification pending. No measured fit.

HC-001 / INS-001 manufacture and assembly datums





Clamp boresØ0.95±0.05; accepted wireOD1.05..1.20. SeatZ−2.8; carrier±0.05. Certificates/preload/pull-test pending.

Assembly specification and host requirements

Front carrier:49.2×36.4×4mm;6061-T6/T651 ±0.05mm.

Front motor4×M3 at26±0.2square, depth≥4mm; never use rear screws.

Carrier4×Ø4.1±0.05 at26±0.05; Ø22.15±0.05 locating bore.

Pilot root reliefØ23.4±0.05×0.40±0.05deep from motor mating face.

Motor fasteners4×M3×8 +0.5mm washers; bounded engagement3.2..3.8mm.

Host4×M3 atX±21.6,Y±13;6mm available depth, engage4.5..5.5mm.

Host frontfaceZ−44.2; shaftprojection beyondhost14.9..17.1mm.

PCB4×M2.5×8; PEEKtop0.5/lower1mm; nominal metalengagement4.9mm.

Motor PCBheader:JST B4B-PH-K-S(LF)(SN), C131334; bore0.75±0.05.

MatePHR-4 +4×SPH-002T-P0.5S; AWG24..30, OD0.8..1.5mm.

1 Black A+/OUT1A;2 Green A−/OUT1B;3 Red B+/OUT2A;4 Blue B−/OUT2B.

Belden83004:010100black /005100green /002100red /006100blue.

ExtensionnomOD1.1,PTFE200°C,minstaticbend11mm; modeled12mm.

HAR-002 factorylead OD/AWG/material unknown: inspectbeforecutting.

Individually insulated/staggered lapsplices in bounded2.8×8×2.8 envelopes.

Spliceinsulation candidateRT-375-1/8-X; certificate/process qualification pending.

Tail projection≤0.8,fillet≤0.5 belowPCB; qualified temperature-rated tailcaps.

INS-001Nomex4100.51; ≥150°C materialcertificate required.

HC-001PEEK/VMQ50±5ShoreA,2×M2.5×10;4mm nominalmetalengagement.

Headerwithdrawal≥10mm, grippinghousing; deenergize/discharge before unplug.

PD Tensility10-06137; DATA Tensility10-06139; maximumbody12.5×32.5×7.6.

USBpitch14mm; worstplacementbodyseparation1.3mm; reserveR30 bend.

R30 is hostspace, not manufacturer-qualified cablebend/fatigue radius.

Complete reservedbbox(mm): [-24.602, -70.082, -60.211, 24.602, 87.399, 42.559]

Minimum hostenvelope(mm): [49.204, 157.48, 102.771]